Designing an Interactive Educational Framework for LZSCC.111

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Abstract

Students on LZSCC.111 Software Development learn to program in C, where program state is hidden and mistakes often show up only as a wrong number or a crash. This project will build a framework of interactive C exercises that increase in difficulty. In each exercise the student implements a single function. Provided scaffolding handles input and output, and a Python visualiser shows what the code does step by step, such as the bars of an array being searched or a robot moving through a maze. Student code runs inside a C harness that records each step as an event trace and catches crashes, infinite loops and out-of-bounds access. The framework will be evaluated technically and through a small study with students, comparing the exercises with and without the visualiser. The expected outcome is a reusable set of tasks for LZSCC.111, along with indicative evidence on whether visual feedback improves novice understanding and engagement.

1 Introduction

LZSCC.111 introduces students to programming in C. For a beginner, C gives little visible feedback. A loop that searches an array or a function that moves a robot produces at most a line of printed output, and the student cannot see how the program reached its result. Reading input, parsing it and formatting output also take effort unrelated to the concept being taught, so beginners spend their time on scanf and printf instead of the algorithm. Research on algorithm visualisation has found that showing execution helps most when students interact with the visualisation rather than only watch it. This project applies that idea to LZSCC.111. Students write only the core logic of each task in C, and a Python visualiser shows its effect. The rest of this proposal is organised as follows. Section 2 reviews related work and the gap this project addresses. Section 3 sets out the aims, objectives and research questions, followed by the methodology. Section 4 describes the programme of work, Section 5 lists the resources required, and the references follow.

2 Background and Related Work

Show where your project fits into existing work: similar projects, tools, systems, and papers **example-article**, **example-inproceedings**. *This is an integrated synthesis, not a list of who did what. For each source, explain why it is relevant to your project, what its limitations are ('however, . . . '), and how your project addresses them ('therefore, . . . ').*

Organise the section thematically, chronologically, methodologically, or as a funnel (general background → closely related work). End with the scope of your project: what exactly you will do, and what not.

3 Proposed Project

3.1 Aims and Objectives

State the overall aim in one sentence. Then list the objectives. Every objective must be measurable: how will you (and the examiners) know that you achieved it?

The aim of this project is to . . .

O1 Objective one – *e.g. 'Implement a route planner that finds optimal routes on the map of Leipzig in under one second.'*

O2 Objective two – . . .

O3 Objective three – . . .

3.2 Methodology

Describe in detail how you will achieve each objective: development process, architecture and technologies, algorithms, datasets, experimental setup, user studies, . . .

3.3 Evaluation

Your plan for analysis: how will you evaluate the results? Name concrete metrics, baselines, benchmarks, test suites, or study designs, and say how they relate to the objectives in Section 3.1.

3.4 Ethics

Only if your project involves people (user studies, surveys, interviews) or personal data: explain what you will collect, how you will protect participants and data, and when you will apply for ethics approval. Otherwise delete this subsection.

4 Programme of Work

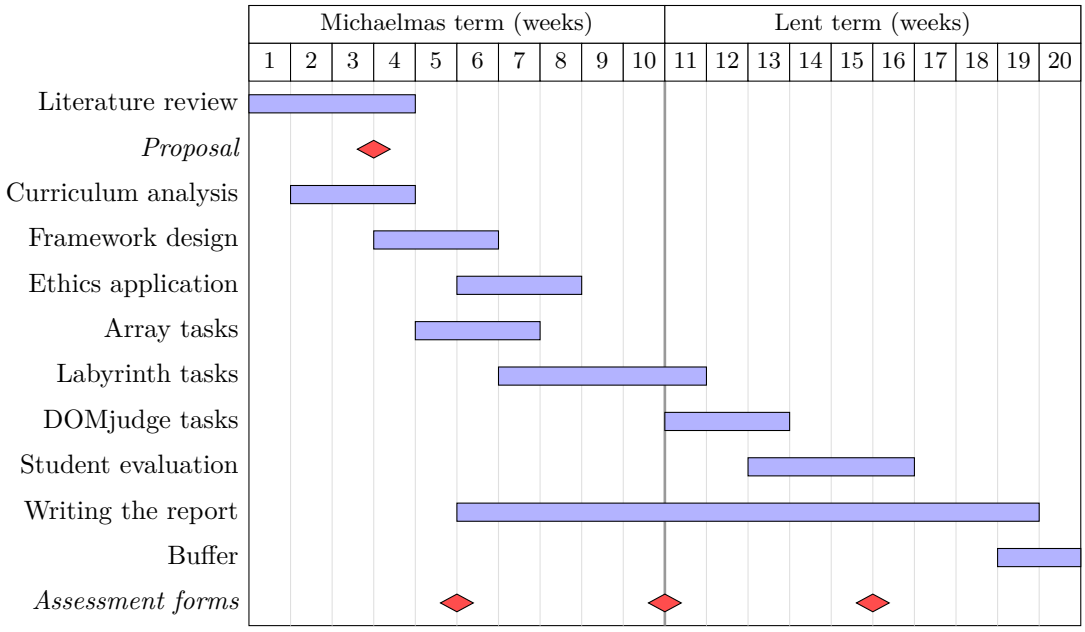


Figure 1: Project schedule. There is a four-week break between weeks 10 and 11.

5 Resources Required

- **Software:** gcc/clang, Python, pygame and Git are all free and open-source, and I will install them myself. A free personal GitHub account (username: florea2005). Overleaf: a free account, or access through Lancaster's email.

- **Hardware:** My personal laptop will be used for development, testing and writing the report. No special hardware or GPU is needed. Lancaster University Leipzig's lab machines will be used during testing to confirm that the tasks run in the real LZSCC.111 environment.

6 Risks and Contingency

What could go wrong, and what will you do then? Think about risks specific to your project.

Table 1: Risks and mitigation.

Risk	Likelihood	Impact	Mitigation
Dataset not available in time	medium	high	Use synthetic data first, contact provider early
Data loss	low	high	Version control, regular backups
Illness or other absence	low	medium	Buffer time in the schedule